Lec27

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Product of array except self.

prefix suffix

→ arr = [1, 2, 3, 4]

result = [24, 12, 8, 6]

prefix = [1, 1, 2, 6]

suffix = [24 | 12 | 4 | 1]

2×4
1×12

result[0] = 2×3×4
result[1] = 1×3×4
result[2] = 1×2×4
result[3] = 1×2×3

1×1
1×2
2×3

1×4
4×3
12×2

1×24
1×12
2×4
6×1

```
int prefix = new int[arr.length]
prefix[0] = 1
for(i=1; i<n; i++){
    prefix[i] = prefix[i-1] x arr[i-1]
}
int suffix[] = new int[arr.length]
suffix[n-1] = 1
for(i=n-2; i>=0; i--){
    suffix[i] = suffix[i+1] * arr[i+1]
}
int result = new int[n]
for(i=0; i<n; i++){
    result[i] = prefix[i] * suffix[i]
}
return result.
```

Trappin rain water

→ arr = [5, 2, 1, 4, 7]

0 1 2 3 4

Left = [5 | 5 | 5 | 5 | 7]

Right = [7 | 7 | 7 | 7 | 7]

8

min(Left[i], Right[i]) − arr[i] = min(5,7) = 5 − 5 = 0
min(Left[1], Right[1]) − arr[1] = min(5,7) = 5 − 2 = 3
min(Left[2], Right[2]) − arr[2] = min(5,7) = 5 − 1 = 4
min(Left[3], Right[3]) − arr[3] = min(5,7) = 5 − 4 = 1
min(Left[4], Right[4]) − arr[4] = min(7,7) = 7 − 7 = 0

100 → manthan

101 → Gaurav

$\min(\text{Left}[3], \text{Right}[4]) - arr[4] = \min(7,7) = 7 - 7 = 0$

$\min(\text{Left}[4], \text{Right}[4]) - arr[4] = \min(7,7) = 7 - 7 = 0$

101 → Gausian

$arr = [5, 2, 1, 4, 7]$

$\uparrow \; i \; \; i$

$\text{Left} =$

5	5			

$i \; i \; \; i$

$\text{Left}[i] = \max(\text{Left}[i-1], arr[i]) = (5, 2)$

$= (5, 1)$